OPTIMIZATION TECHNIQUES · GROUP PROJECT

Multi-Objective College Timetable Optimization

Feasible schedules. Explicit trade-offs. Reproducible evidence.

168

held-out runs

3.9–9.0%

primary mean HV gains

0

invalid exported schedules

A scheduling decision with three interests

Students

Attend every required session.

Avoid internal idle gaps.

Faculty

No double bookings.

Compact and balanced teaching.

College

Use compatible rooms.

Match class sizes to seats.

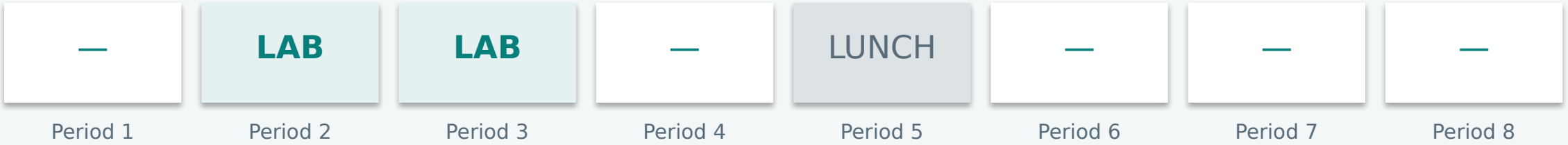
- Fixed inputs: demand, duration, cohorts, rooms, eligibility, availability and limits.
- Decisions: one start + room per event; one eligible teacher per offering.

All weekly sessions of an offering share the same teacher.

Evidence: problem/README.md; formulation/README.md, Sections 1–2

Hard constraints define a usable timetable

- Every session exactly once; consistent eligible offering teachers.
- No room, faculty or student-group clash in any occupied period.
- Sufficient capacity, exact room type and full-block availability.
- Daily/weekly teaching limits; distinct teaching days when requested.



A lab is one contiguous block: it cannot cross lunch or a day boundary.

Evidence: formulation/README.md, C1–C8; evaluation.py

Three objectives, all minimized

f1 Idle gaps

$$f_1 = \sum_{g,d} \text{gap}_{g,d} + \tfrac{1}{2} \sum_{f,d} \text{gap}_{f,d}$$

Internal free teaching periods; mandatory lunch excluded.

f2 Room wastage

$$f_2 = \frac{1}{\sum_e \ell_e} \sum_e \ell_e (1 - n_e / c_{r(e)})$$

Duration-weighted unused seat fraction during scheduled use.

f3 Workload imbalance

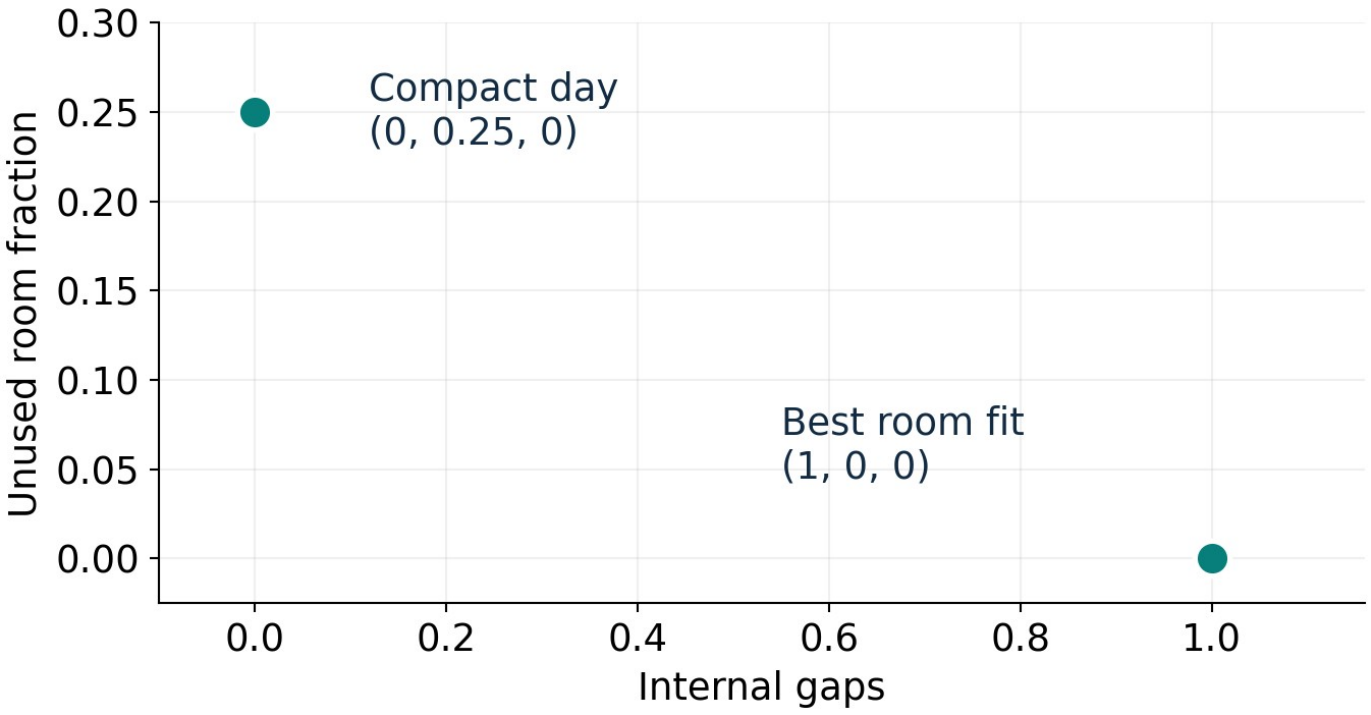
$$f_3 = \sqrt{\frac{1}{|F|} \sum_f (L_f / H_f - \bar{\rho})^2}$$

Population SD of teaching load divided by weekly allowance.

Teacher choice makes workload optimizable; changing only times would not.

Evidence: formulation/README.md, Section 4

Conflict is demonstrated, not assumed



- Small room: available at separated periods 0 and 2.
- Larger room: available at periods 1 and 3.
- Zero wastage forces a gap; zero gap requires extra seats.

64 complete candidate evaluations certify this tiny global front.

Evidence: results/exact_micro/room/exact_front.json; formulation Section 5

Synthetic data with known feasibility

Preset	Events	Cohorts	Rooms	Faculty
Small	40	2	5	6
Medium	150	8	12	20
Large	400	20	25	50

- Five days × eight periods; fixed lunch; two-period full-cohort labs.
- Deterministic seeds; configurable demand, resources and availability.
- A separate feasible certificate is generated and checked, then withheld from solvers.

Planted feasibility separates algorithm failure from impossibility—and biases sampling.

Evidence: data/README.md; generation.py

The final method targets an observed weakness

24 feasible constructive seeds › **Rank/crowding parent selection** › **Feasible mutation + faculty search** › **Counted full validation** › **Elitist P + Q survival**

- Strong reference: repeated DSatur-style greedy construction + a feasible Pareto archive.
- Historical enhanced NSGA-II: more than 90% of raw offspring required repair.
- Final method: fully feasible initialization and resource-aware mutation.
- Retained: true rank/crowding tournaments and elitist parent–offspring survival.

Mutation-only memetic NSGA-II: destructive crossover is deliberately omitted.

Evidence: docs/enhancement_study.md; memetic.py; historical benchmark

Domain knowledge, with bounded cost

Five resource-aware moves

- Move a session to a feasible block
- Change to a compatible fitting room
- Swap event blocks safely
- Reinsert an offering with a new teacher
- Exchange cross-qualified offering teachers

Structured faculty reassignment

- Reassign up to 8 whole offerings at fixed times/rooms.
- Branch-and-bound capped at 12,000 nodes per call.
- Equal allowances: smaller squared loads means smaller true workload SD.
- Recheck all three objectives and every hard constraint.

Partial checks and search nodes consume time; full evaluator calls are not equal CPU work.

Evidence: docs/feasible_variation.md; feasible.py; faculty_search.py

Frozen, held-out validation

Size	Instance seeds	Solver seeds	Full-eval cap	Wall allocation
Small regression	1010	0, 1, 2	720	2.5 s
Medium primary	1010 / 1020 / 1030	0, 1, 2	360	6 s
Large primary	1010 / 1020 / 1030	0, 1, 2	240	16 s

- Four methods: greedy, original enhanced NSGA-II, final memetic NSGA-II and scalar-local ablation.
- Two separate budget protocols; 168 completed runs. Development seeds 9000+ stay separate.
- Unchanged presets, objectives, normalization and HV reference (1.1, 1.1, 1.1).

Evidence: experiments/enhancement_validation.json; manifest.json

Fair budgets and verified cutoff evidence

201
pre-freeze tests passed

168 / 168
declared runs verified

3,295
valid archived schedules

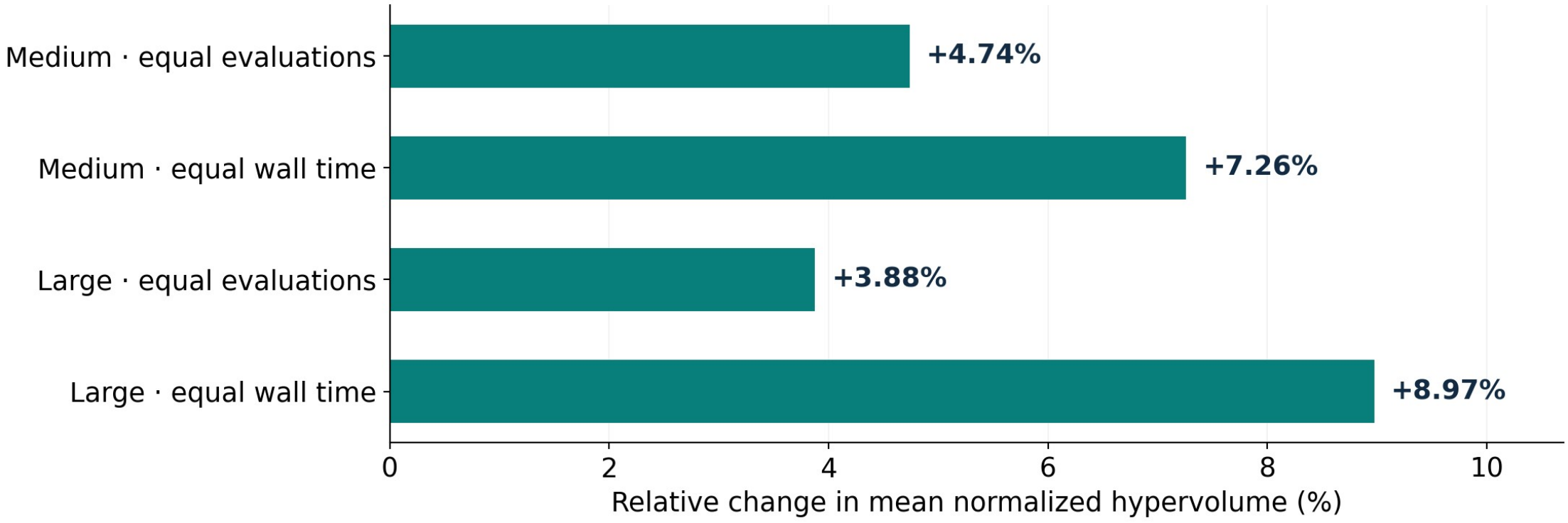
9
late evaluations excluded

- Charge initialization, duplicates, repair and rejected full local proposals.
- Time preprocessing, partial checks, native bookkeeping and common observation.
- Reset constructor caches; run methods sequentially in rotated order.
- Only complete objective results available by the deadline receive quality credit.

Seven separate feasibility certificates checked; no empty archives in this validation.

Evidence: verification.json; comparison.py; tests_before_validation.txt

The final method improves the primary comparisons

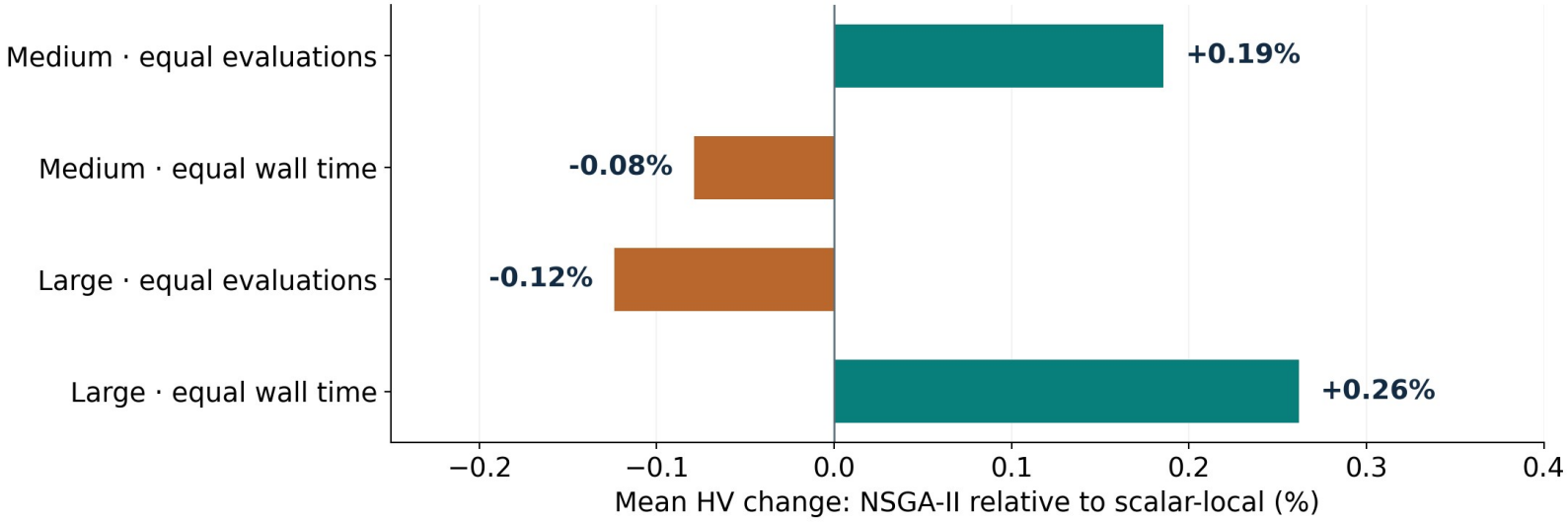


- 9 paired wins out of 9 in each medium/large condition.
- Equal-evaluation mean runtime: 2.93 s vs 20.22 s (medium); 8.04 s vs 51.67 s (large).
- Observed synthetic gains, not universal superiority or global Pareto optimality.

Evidence: summary.json; report/enhancement_report.md

Keep contrary evidence and the ablation

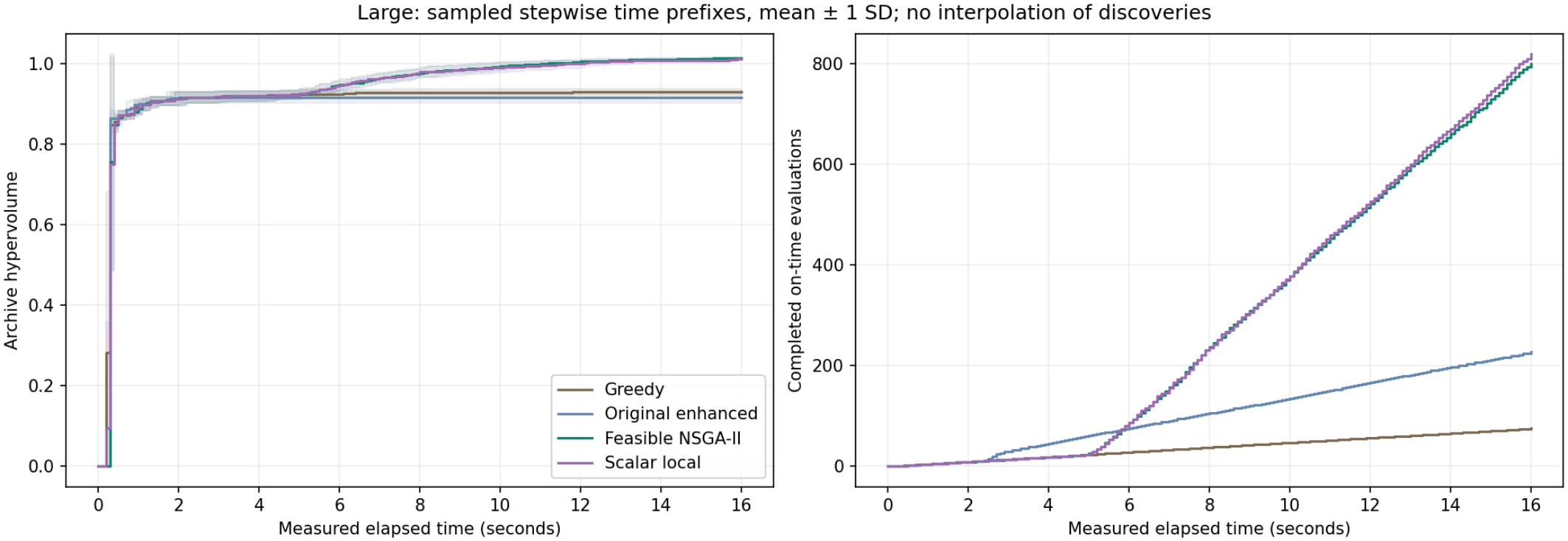
Small regression	Greedy HV	Memetic HV	Change
Equal evaluations	0.934747	0.930290	-0.48%
Equal wall time	0.929781	0.930290	+0.05%



Scalar-local search is competitive; NSGA-II selection has no consistent demonstrated advantage.

Evidence: summary.json: memetic_vs_ablation_local; enhancement_report.md

Convergence follows real candidate discoveries



- Quality and evaluation-count curves use real completion timestamps.
- Shading: descriptive one-SD variability across solver runs, not independent-instance confidence intervals.
- Three independent primary instances per size: minimum two-sided instance sign-test p = 0.25.

Evidence: figures/enhancement_validation/large_anytime.png; instance-level comparisons

A complete baseline timetable: balanced workload

synthetic-medium-1010: group G01
Periods are categorical; lunch is blocked

	Period 1	Period 2	Period 3	Period 4	Period 5	Period 6	Period 7	Period 8
Monday	G01-T4 R01 / F20	G01-T1 R01 / F17	G01-T5 R05 / F15	G01-T2 R01 / F04	LUNCH	—	—	—
Tuesday	—	—	—	—	LUNCH	—	G01-T3 R05 / F16	G01-T5 R01 / F15
Wednesday	—	—	—	G01-T2 R05 / F04	LUNCH	G01-T4 R01 / F20	G01-L1 R09 / F16	G01-L1 (cont.) R09 / F16
Thursday	—	G01-T4 R01 / F20	G01-T6 R05 / F11	G01-T3 R01 / F16	LUNCH	G01-T1 R01 / F17	G01-T5 R01 / F15	G01-T2 R05 / F04
Friday	—	—	G01-L2 R09 / F05	G01-L2 (cont.) R09 / F05	LUNCH	G01-T1 R01 / F17	G01-T3 R01 / F16	G01-T6 R01 / F11

3.0
global weighted gaps

0.192972
unused-seat fraction

0.050249
relative-load SD

6–11
faculty weekly load range

Repeated greedy · instance 1010 · seed 0 · 6-second protocol · same SD / gaps / room preference.

Evidence: representatives.json: medium_1010 / greedy / balanced_workload

The matched final timetable improves this choice

synthetic-medium-1010: group G01
Periods are categorical; lunch is blocked

	Period 1	Period 2	Period 3	Period 4	Period 5	Period 6	Period 7	Period 8
Monday	—	G01-T2 R05 / F08	G01-T5 R01 / F14	G01-T3 R01 / F16	LUNCH	G01-T4 R05 / F20	—	—
Tuesday	G01-T1 R05 / F05	G01-L1 R09 / F18	G01-L1 (cont.) R09 / F18	G01-T5 R05 / F14	LUNCH	G01-T3 R05 / F16	G01-T4 R05 / F20	—
Wednesday	—	G01-T6 R01 / F04	G01-T3 R05 / F16	G01-T1 R05 / F05	LUNCH	—	—	—
Thursday	—	G01-T5 R05 / F14	G01-T1 R05 / F05	G01-T6 R05 / F04	LUNCH	G01-T2 R05 / F08	—	—
Friday	G01-T4 R01 / F20	G01-T2 R01 / F08	G01-L2 R09 / F09	G01-L2 (cont.) R09 / F09	LUNCH	—	—	—

2.0
global weighted gaps

0.191165
unused-seat fraction

0.022913
relative-load SD

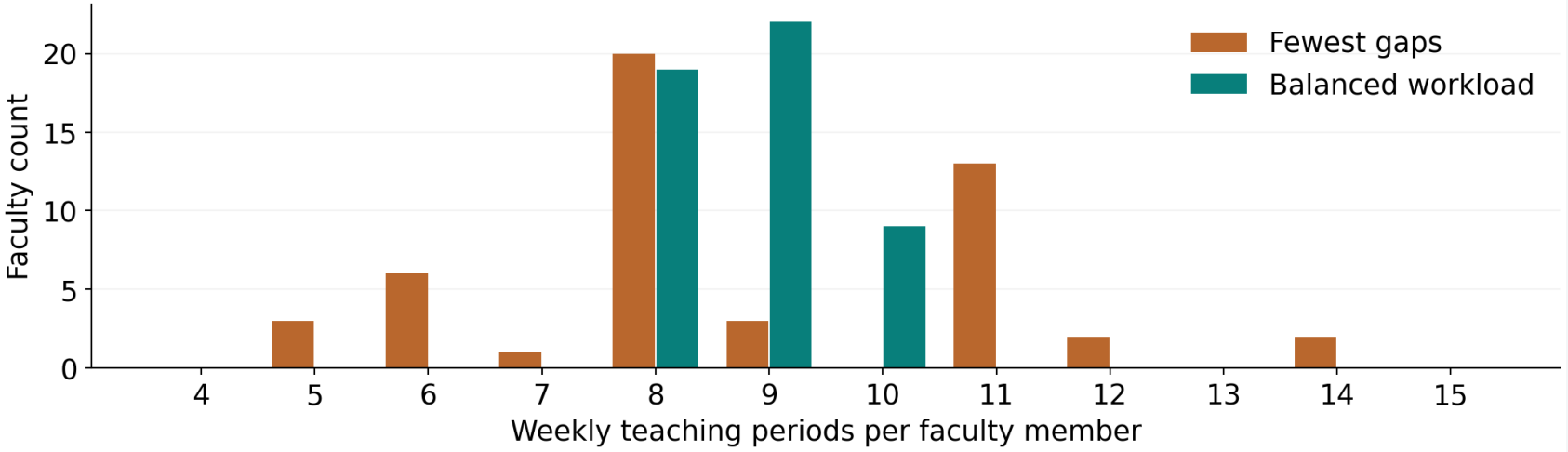
8–9
faculty weekly load range

Final memetic method · instance 1010 · seed 0 · 6-second protocol · same SD / gaps / room preference.

Evidence: representatives.json: medium_1010 / memetic / balanced_workload

Better balance can cost real idle time

Memetic choice	Weighted gaps	Unused seats	Load SD	Student / faculty gaps
Fewest gaps	1.5	0.208333	0.110454	0 / 3
Balanced workload	23.5	0.205985	0.036056	8 / 31



The coordinator chooses the compromise; hypervolume is not an administrative utility function.

Evidence: representatives.json: large_1010 / memetic / both preference selections

Know which guarantees apply

Established

- Every exported schedule satisfies modeled hard constraints.
- Tiny exhaustive fronts are globally certified.
- small_1010 attains the singleton global ideal (0, 14/55, √(1/450)).
- Medium schedules attain a workload-only relaxation bound.

Not established

- Global medium/large Pareto optimality.
- Consistent advantage over scalar-local search.
- Real-college or ITC benchmark performance.
- Broad statistical superiority from three primary instances.

A discovered non-dominated archive is not a global Pareto certificate.

Evidence: small_ideal_certificates.json; formulation/README.md; enhancement_report.md

Reproduce and demonstrate the decision

```
python main.py generate --size small --seed 9002 --output work/presentation_demo/instance.json

python main.py solve work/presentation_demo/instance.json --method memetic --seed 0 --budget 120 --population 12 --
output work/presentation_demo/run --figures work/presentation_demo/figures

python main.py evaluate work/presentation_demo/instance.json
work/presentation_demo/run/timetables/normalized_compromise/schedule.json
```

- Display violation = 0, objective values and the exported weekly grid.
- Keep live examples separate from frozen measurements.
- Next: real coordinator validation, broader instances, and reference/scale sensitivity.

Choose a clash-free timetable by explicit priorities—and keep the evidence behind the choice.

Evidence: presentation/demo.sh; main.py; report/enhancement_report.md

Backup: binary formulation and teacher consistency

$$z_{e,s,r,f} \in \{0,1\}, \quad y_{k,f} \in \{0,1\}$$

$$\sum_{(s,r,f) \in A_e} z_{e,s,r,f} = 1, \quad \sum_{f \in F_k} y_{k,f} = 1$$

$$\sum_{(s,r) : (s,r,f) \in A_e} z_{e,s,r,f} = y_{k(e),f}$$

- A_e filters room capacity/type, geometry, lunch and full-block unavailability.
- Resource occupancy ≤ 1 for every room, teacher and cohort in every period.
- Weekly offering loads and daily occupied periods obey faculty limits.

Discrete constrained feasibility is binary linear; workload SD is nonlinear.

Evidence: formulation/README.md, C1–C8; exact.py

Backup: the small ideal is a valid exact certificate

- 44 teaching periods; 6 faculty; common allowance 20.
- Indivisible offering loads: twelve 3-period and four 2-period offerings.
- Exact partition relaxation: min squared loads = 328 → variance ≥ 1/450.
- Independent bounds: gaps ≥ 0 and unused-seat fraction ≥ 14/55.
- A feasible timetable attains every bound: loads [6, 6, 8, 8, 8, 8].

Simultaneous attainment proves a singleton ideal; it does not prove reliable attainment across runs.

Evidence: small_ideal_certificates.json; experiments/certify_small_ideal.py

Backup: references and viva checkpoints

- Deb et al. (2002), NSGA-II. IEEE TEC. DOI: 10.1109/4235.996017.
- Brélaz (1979), New Methods to Color the Vertices of a Graph. DOI: 10.1145/359094.359101.
- Schaerf (1999), A Survey of Automated Timetabling. Artificial Intelligence Review.
- Zitzler et al. (2003), Performance Assessment of Multiobjective Optimizers. IEEE TEC.
- Google OR-Tools CP-SAT documentation; professor assignment and supplied proposal.

Rebuild: python presentation/build_presentation.py · Evidence values/hashes: presentation/evidence.json

Evidence: literature/README.md; docs/reference_review.md; manifest.json